OMNISERVICE CENTER

ONE SMART PLACE FOR EVERY SERVICE.

SERVICE SEEKER	OMNI INTELLIGENCE	SERVICE PROVIDER
Discover • Request • Review	Match • Rank • Connect	Offer • Quote • Deliver

A smart digital service ecosystem for discovering, connecting, managing, and accessing services.

SCIENCE FAIR COMPETITION PROPOSAL

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2019 E.C.

01 • INTRODUCTION

Services are everywhere—but finding the right service is not always simple. A person may need a technician, tutor, transport provider, healthcare-related service, consultant, agricultural service, event provider, or community organization and still have to search across disconnected channels.

OMNISERVICE CENTER was created around one simple observation: the service may already exist, and the provider may already exist, but the connection between them can be fragmented.

The project proposes a unified digital ecosystem where people can discover services, describe what they need, find suitable providers, submit requests, receive quotations, manage service interactions, and provide reviews.

The prototype is designed as a working web application rather than a collection of static screens. It combines a frontend, backend API, persistent prototype storage, authentication, intelligent service matching, trust logic, mapping, payment sandbox, and administration.

At its heart is a simple principle:

“The service you need may already exist. The provider you need may already exist. The challenge is connecting the two.”

This proposal presents the problem, concept, purpose, system features, expected impact, technical approach, current prototype, limitations, and future direction of OMNISERVICE CENTER.

PROJECT AT A GLANCE

12	3	1
Service sectors	Core roles	Unified ecosystem

02 • OVERVIEW & PURPOSE

OMNISERVICE CENTER is an Ethiopia-focused service ecosystem connecting Service Seekers, Service Providers, companies and organizations through one digital platform. It is designed around the complete service journey rather than only service search.

The platform has three main perspectives:

SERVICE SEEKER	Find services, use OMNI, request help, compare provider information, receive quotations, accept services, and review completed interactions.
SERVICE PROVIDER	Create a service presence, receive requests, respond with quotations, manage jobs, and build service history and reputation.
ADMINISTRATION	Verify providers, manage users, oversee activity, generate reports, and provide governance for the ecosystem.

PURPOSE

The purpose of the project is to demonstrate how software can reduce fragmentation in service discovery and coordination. The prototype aims to make service access more organized, transparent, and manageable while also giving providers a structured digital channel to reach potential customers.

GENERAL OBJECTIVE

To design and develop a functional digital service ecosystem that connects service seekers with suitable service providers through intelligent matching, structured workflows, trust mechanisms, location services, and administrative control.

DESIGN PHILOSOPHY

Simple for the seeker. Describe the need. Useful for the provider. Receive the opportunity. Responsible for the platform. Verify, monitor, and govern. Honest about the prototype. Clearly distinguish current capabilities from future production features.

03 • THE PROBLEM & IMPACT

The project addresses a coordination gap between service demand and service supply.

COMMON FRICTIONS

Fragmented discovery	Users may search through social media, calls, websites, personal contacts, and advertisements.
Provider visibility	Capable local providers can be difficult for potential customers to discover.
Trust uncertainty	A name or phone number does not necessarily provide enough structured information for comparison.
Disconnected workflow	Discovery, communication, quotation, booking, payment, and review can happen in separate places.
Limited oversight	A growing ecosystem needs verification, monitoring, reporting, and administrative control.

EXPECTED IMPACT

For service seekers, the platform could simplify discovery and provide structured information for decision-making. For providers, it could increase digital visibility and create a consistent way to receive and manage requests. For organizations, it could create a single environment for discovering and coordinating services.

The wider concept is digital coordination: connecting people who need services with people who provide them.

IMPACT CHAIN

Need → Discovery → Matching → Connection → Service → Feedback → Trust

04 • FEATURES & SERVICE ECOSYSTEM

OMNISERVICE CENTER organizes a broad range of services while keeping the user journey consistent.

12 SERVICE SECTORS

01 IT & Technology	02 Transportation	03 Telecommunications
04 Government Services	05 Education	06 Healthcare
07 Real Estate	08 Business & Consulting	09 Shopping & Supermarkets
10 Agriculture	11 Events & Media	12 NGOs & Community Services

CORE FEATURES

✓ Role-based access	✓ Service discovery
✓ OMNI intelligent matching	✓ Provider profiles
✓ Service requests	✓ Provider quotations
✓ Booking/job lifecycle	✓ Reviews & trust information
✓ Leaflet map	✓ Payment sandbox
✓ Provider verification	✓ Admin dashboard & reports

05 • HOW IT WORKS & OMNI

The strongest part of the concept is the end-to-end journey. OMNISERVICE CENTER is designed to move from a service need to a completed service interaction.

THE SERVICE JOURNEY

01 DISCOVER → 02 DESCRIBE → 03 MATCH → 04 SELECT → 05 REQUEST → 06 QUOTE → 07 ACCEPT → 08 COMPLETE → 09 REVIEW

OMNI INTELLIGENT MATCHING

OMNI is the platform's specialized service-matching engine. It is important to distinguish the current implementation from a general-purpose generative AI model. OMNI is not a trained large language model; it is a rule- and scoring-based natural-language matching system designed around the platform's service data.

A typical request can be processed through:

User text → normalize input → identify service intent → identify city/location → detect timing or urgency → filter verified providers → calculate matching factors → rank providers → return suitable matches.

The prototype uses service synonym mappings, city aliases, timing patterns, trust information, availability, price level, review-related information, and geographic distance. Geographic distance can be calculated using the Haversine formula.

WHY THIS MATTERS

Instead of forcing the user to know exactly which category or provider to search for, the system allows the user to begin with the problem they want solved.

Example: “I need computer repair in Gondar.” OMNI can identify the service intent and location, then search the verified provider pool for suitable matches.

06 • TECHNOLOGY, TRUST & SECURITY

The prototype is implemented as a full web application with a clear separation between interface, API, business logic, and data.

TECHNICAL ARCHITECTURE

USER → FRONTEND → EXPRESS API → BUSINESS LOGIC → DATA → API RESPONSE → FRONTEND

Frontend	HTML, CSS, JavaScript
Backend	Node.js + Express
Authentication	JWT + bcrypt password hashing
Data	Persistent JSON-based datastore abstraction
Maps	Leaflet + OpenStreetMap tiles
Intelligence	OMNI matching + ranking logic
Trust	Prototype trust-score calculation
Payments	Sandbox workflow and validation

TRUST & VERIFICATION

New providers can enter a pending state and require administrative verification. The current OMNI matching workflow considers verified providers. The prototype trust calculation combines review rating, completion rate, response time, and job volume, with cancellations treated negatively.

SECURITY

The prototype uses bcrypt for password hashing, JWT authentication, and backend role-based authorization. A production deployment would require HTTPS, stronger secret management, rate limiting, auditing, monitoring, and a production-grade database.

07 • INNOVATION, IMPACT & SUSTAINABILITY

The central innovation is not simply a directory of providers. It is the attempt to connect the entire service journey in one ecosystem.

DISCOVERY + MATCHING + REQUEST + QUOTATION + BOOKING + REVIEW + TRUST + LOCATION + ADMINISTRATION

This creates a platform model in which the user can move from “I need something” to “I found and completed the service” without treating every stage as a separate system.

POTENTIAL IMPACT

For seekers	More organized discovery, clearer provider information, structured requests, and a connected service journey.
For providers	Improved digital visibility, structured incoming requests, quotation tools, and reputation building.
For communities	A possible digital bridge between existing local services and people who need them.
For organizations	A centralized environment for discovering and coordinating diverse service needs.

SUSTAINABILITY / BUSINESS MODEL

Future revenue possibilities include provider premium accounts, clearly identified promotional placements, advertising, selected transaction/service fees, and organization subscriptions. These are future possibilities rather than current revenue claims.

VISION

Start with a working ecosystem concept. Expand verified providers. Strengthen infrastructure. Improve matching intelligence. Add secure production payments. Expand geographic coverage. Build mobile applications.

08 • CURRENT PROTOTYPE & ROADMAP

CURRENTLY DEMONSTRATED

The current prototype supports an integrated demonstration flow including authentication, role-based experiences, OMNI matching, provider selection, service requests, quotations, booking/job progression, reviews, trust information, mapping, payment sandbox behavior, and administration.

It is therefore more than a visual mockup: the components interact through a working frontend and backend.

IMPORTANT PROTOTYPE LIMITATIONS

Data	Provider information used for demonstration is prototype/demo data.
Storage	JSON storage is appropriate for a portable prototype but not for millions of production records.
Payments	Payment is sandbox/demo only; no real money is processed.
AI	OMNI is specialized rule/scoring-based matching, not a trained LLM.
Verification	Full real-world identity/business verification is future work.
Scale	Production-scale caching, indexing, monitoring, and horizontal scaling are future requirements.

FUTURE ROADMAP

01 Production database & infrastructure • 02 Advanced NLP/ML matching • 03 Verified provider onboarding • 04 Real payment integrations • 05 Mobile applications • 06 Expanded Ethiopian coverage • 07 Fraud prevention, moderation and dispute resolution • 08 Monitoring, analytics and security hardening.

The project intentionally separates what exists today from what could be built next. This keeps the proposal technically credible while showing a clear path for growth.

09 • CONCLUSION & REFERENCES

OMNISERVICE CENTER began with a simple question: if services are everywhere, why should finding and managing them feel so fragmented?

The project responds with a unified ecosystem connecting service seekers, service providers, and administration. Its current prototype demonstrates a service journey from discovery and intelligent matching through request, quotation, booking, completion, review, trust, mapping, and payment workflow.

The project does not claim that the current prototype is already a nationwide commercial platform. Instead, it demonstrates a practical technical foundation and a vision for how such an ecosystem could evolve.

The most important idea is the connection between the people who need services and the people who provide them.

“We are not trying to create more services. We are trying to make the services that already exist easier to find, access, trust, and use.”

OMNISERVICE CENTER

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REFERENCES

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2. Express.js Documentation — official Express web framework documentation.
3. MDN Web Docs — HTML, CSS, JavaScript, browser APIs, and web development references.
4. Leaflet Documentation — interactive web mapping library documentation.
5. OpenStreetMap Foundation — open geographic data and mapping resources.
6. OWASP Foundation — web application security guidance and best practices.
7. IETF RFC 7519 — JSON Web Token specification.
8. JSON.org — JSON data interchange format reference.
9. OMNISERVICE CENTER Project Source Code and Prototype Documentation — project-developed implementation and technical documentation.

PROJECT DECLARATION

OMNISERVICE CENTER is presented as a functional prototype developed for science-fair demonstration. Current capabilities, prototype data, and future development are intentionally distinguished throughout this proposal.

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